CSE422 Lab

Project Report

"Prediction of sterility of tree based on environmental and genetic factors"

Submitted by

Group no: 08

Section: 03

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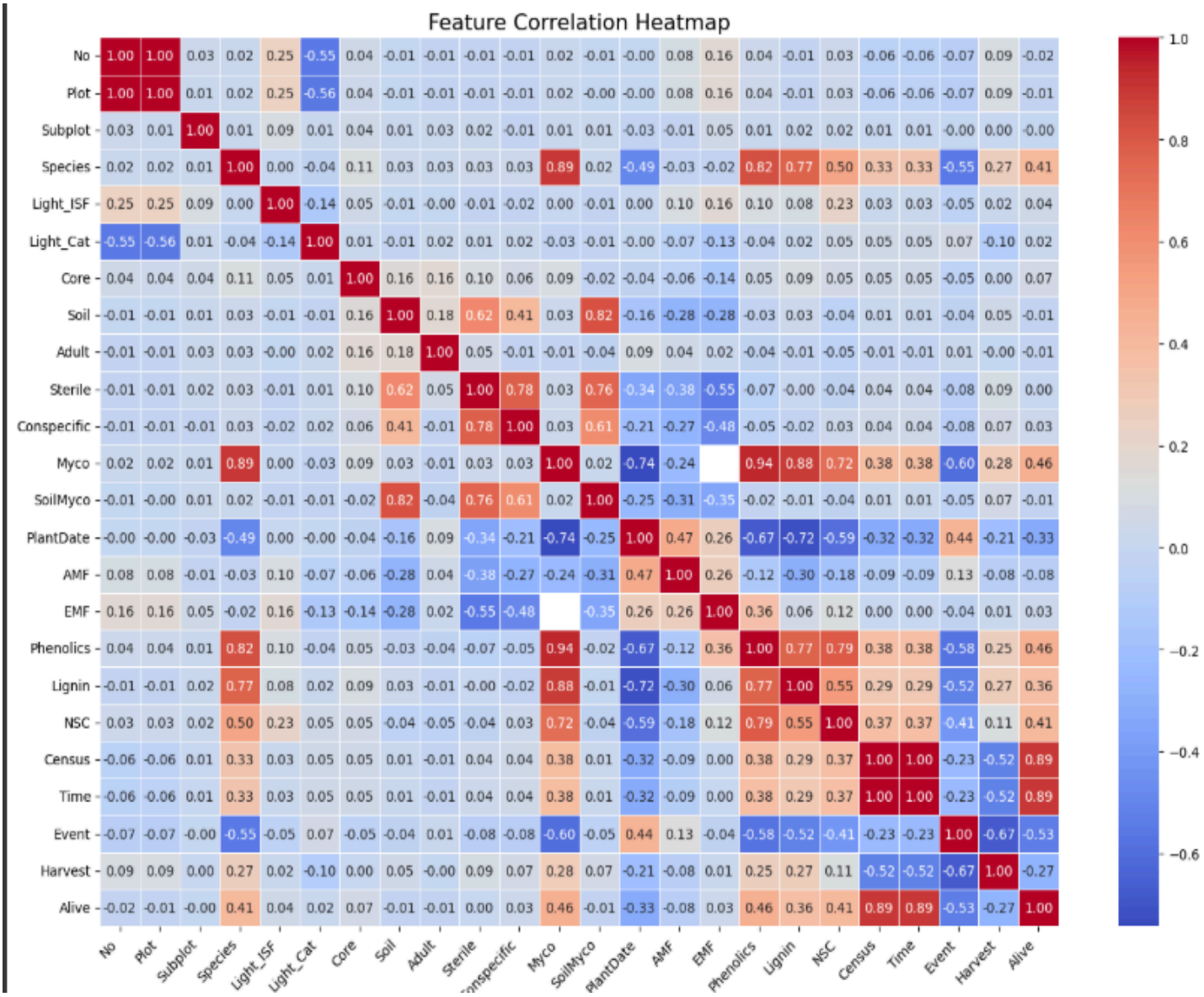
Introduction:

This project aims to predict whether the tree is sterile or non-sterile based on various ecological and biological parameters. The main goal is to assist farmers to make an informed decision by extracting different correlations among those biological factors using machine learning models.

Dataset Description:

Total features: 24
Type of problem: Classification. Because the target column has two distinct labels- sterile and non-sterile.
Data points: 2783
Features: There are 12 numerical features- (No, plot, Light_ISF, Core, AMF, EMF, Phenolics, Lignin, NSC, Census, Time, Event). There are 12 categorical features- (Subplot, Species, Light_Cat, Soil, Adult, Sterile, Conspecific, Myco, SoilMyco, PlantDate, Harvest, Alive)

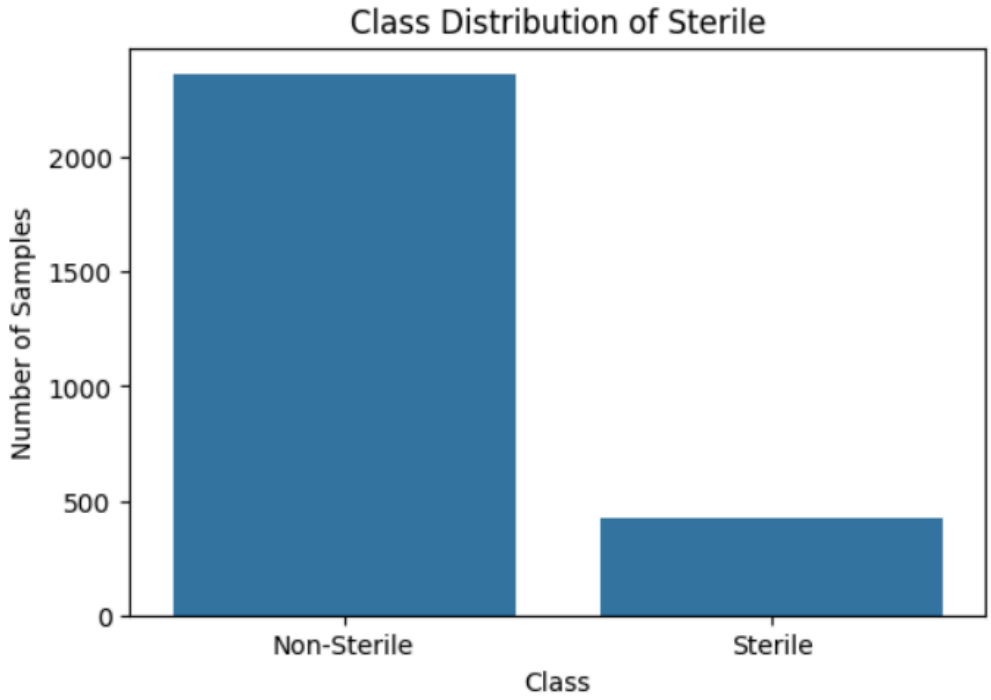
Correlation of all features:



Understanding: Most features show a weak linear correlation with the target, therefore, nonlinear models might perform better.

Imbalanced Dataset:

	outcome	count
0	Non-Sterile	2360
1	Sterile	423

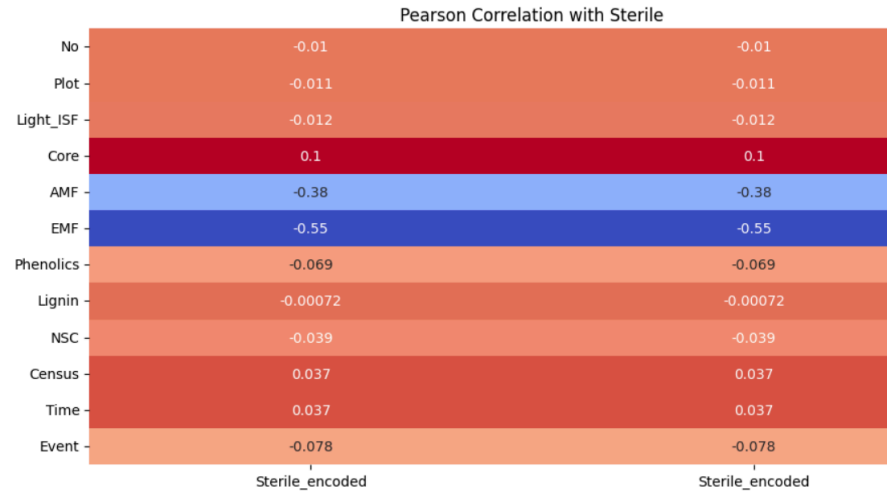


So the target variable has imbalance classes.

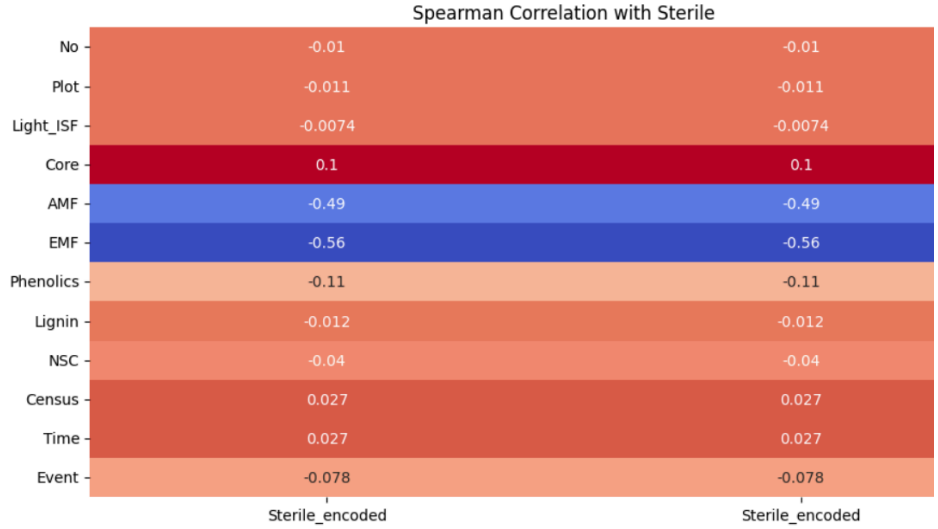
Exploratory Data Analysis:

To better understand the relationship among features and the target label, we performed the following correlation analyses:

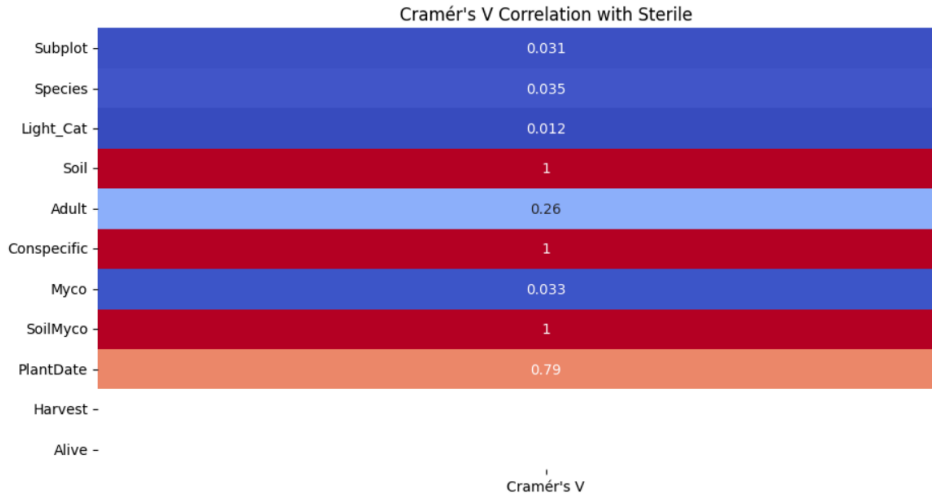
Pearson correlation: The highest negative correlation was found with “EMF”(- .55) and “AMF”(-.38), while “Core” showed the highest positive correlation(0. 1)



Spearman Correlation: Captured monotonic relations. Similar to pearson, EMF (-0.56) and AMF(-0.49) were strongly negatively correlated with sterile. This supports that features consistently influence the outcome in a nonlinear but monotonic way.



Cramers'V correlation: This captures the relationship between categorical variables with target variable-"Sterile".



Key Insights:

- Environmental and biological features(AMF, EMF, Core) influence tree sterility significantly
- Linear correlation is weak overall, but monotonic trends exist.

Dataset Preprocessing:

Several columns, such as EMF, SoilMyco,Event had missing values. Columns with a high percentage of missing values, and irrelevant to prediction were dropped, while categorical missing values were imputed using the mode, numerical values were imputed with the median. Over 10 categorical features were label-encoded to convert them into numerical values suitable for machine learning. To address class imbalance, SMOTE was applied to the training data, generating synthetic samples for better class balance. The numerical features were scaled using MinMaxScaler, ensuring consistency across the dataset.

Dataset Splitting:

Stratified data splitting is used to ensure that the ratio of the classes will be maintained in both train set and test set. 70% of the dataset is used to train the models and 30% is used for testing.

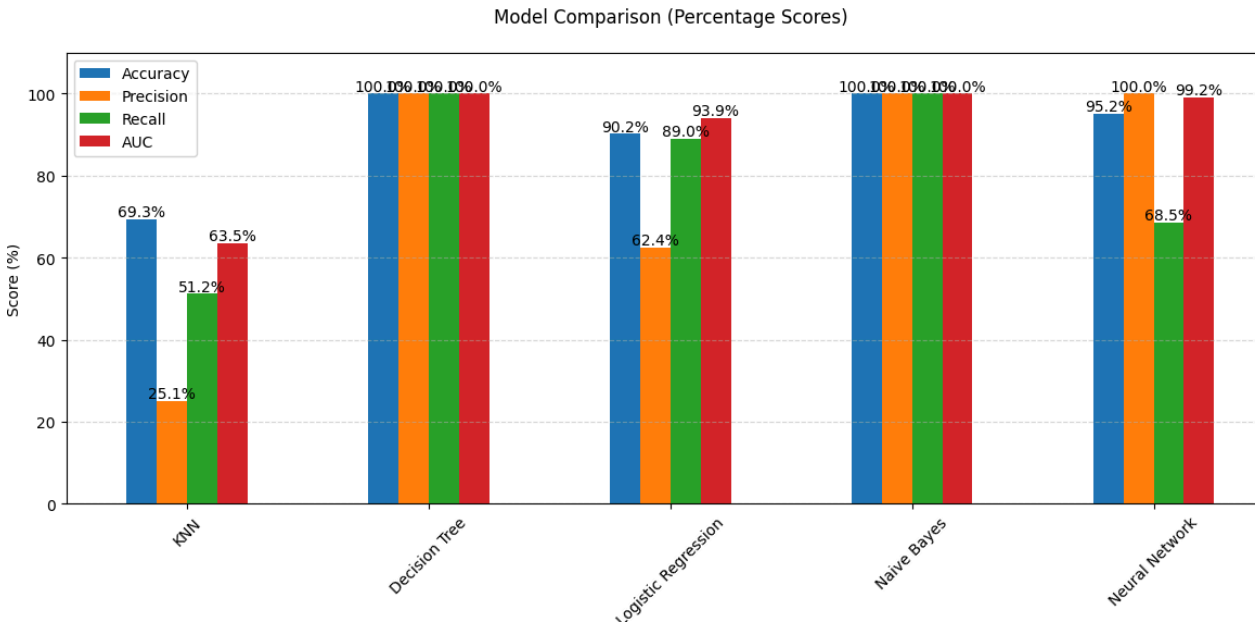
Model training and testing:

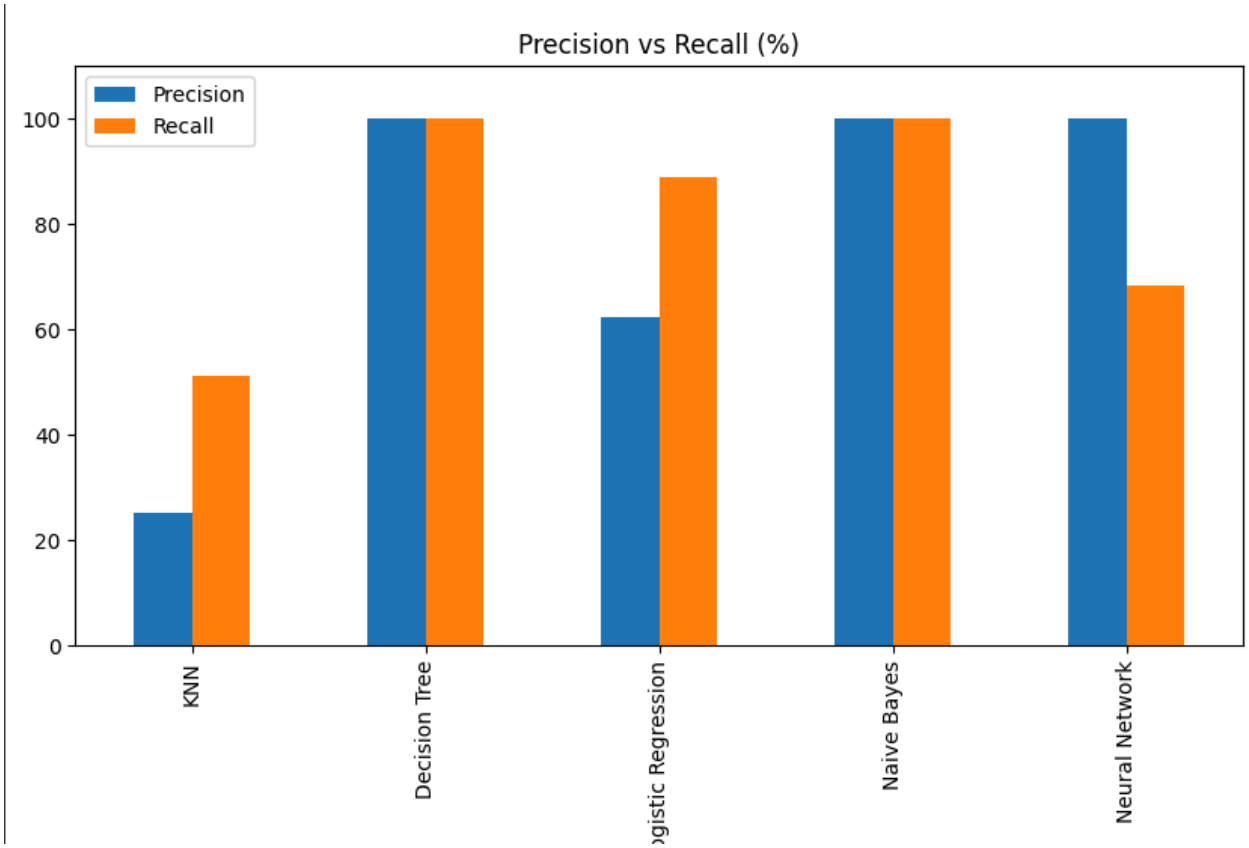
These models are applied-

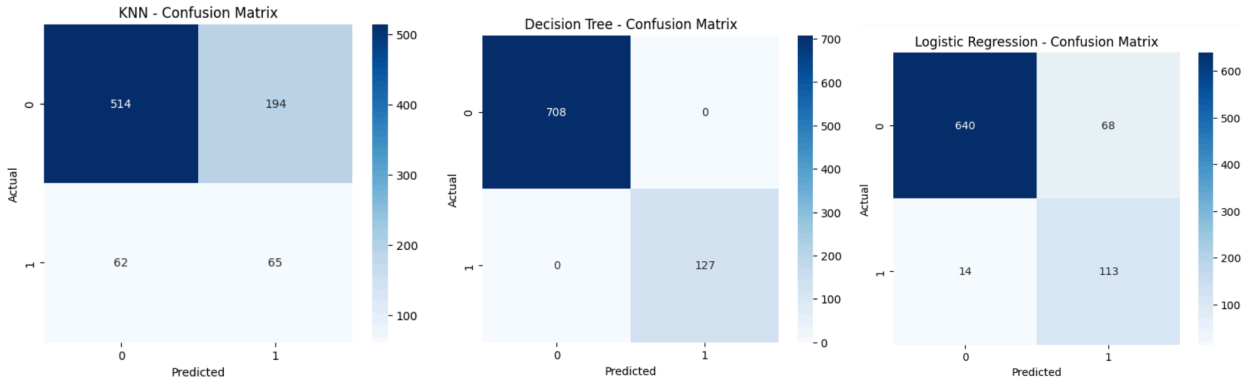
- KNN (K Nearest Neighbours)

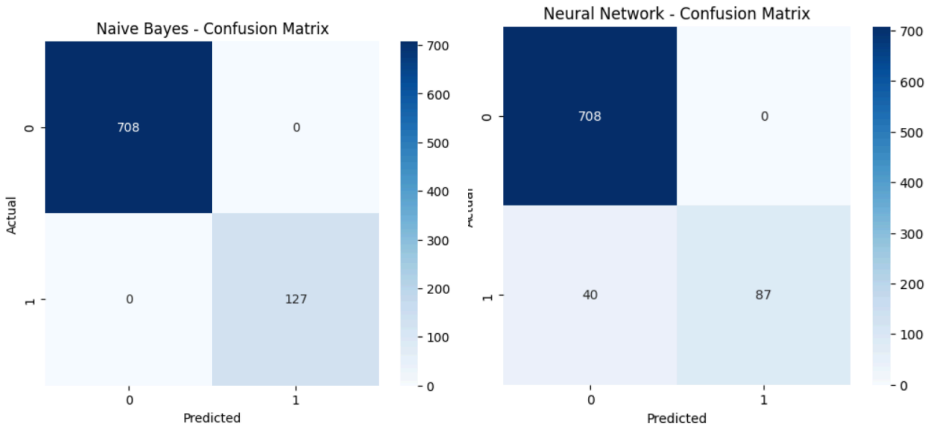
- Decision Tree
- Logistic G=Regression
- Naive Bayes
- Neural network

Comparison Analysis:




Precision vs Recall (%)
Precision
Recall
100
80
60
40
20
0
KNN
Decision Tree
ogistic Regression
Naive Bayes
Neural Network


KNN - Confusion Matrix
514
194
62
65
Actual
Predicted
Decision Tree - Confusion Matrix
708
0
0
127
Logistic Regression - Confusion Matrix
640
68
14
113



From the above metrice, we see **Decision Tree and Naive Bayes** both achieved perfect scores(Accuracy:100%, AUC:1.0), which likely indicates overfitting due to use of SMOTE. Again it suggests that the splitting might be highly perfect, strongly aligned with the assumptions of these models.

Neural Network achieved high accuracy (99.15%), but with a recall of around 70%, showing that it misses some minority class cases. While it's powerful, but its performance is affected by class imbalance and features' high variance.

Logistic regression delivered balanced and robust performance, Accuracy:90.93%, with AUC= .93, precision around 65% and recall around 88-90%. So it indicates that, with high reliability, this model generalizes well and captures linear separability without overly sensitive to synthetic data.

KNN underperformed comparatively ,Accuracy:69.3%, AUC: 0.63, because its performance affected by feature scaling sensitivity and class imbalance.

Conclusion:

Among all tested models, Logistic Regression emerged as the most reliable and generalizable. During model testing we encountered a high class imbalance, which required the use of SMOTE. Then variance skewness in feature distribution, affecting models like Neural Network and KNN. Perfect scores of decision tree and naive bayes are not fully reliable because of overfitting. Taking all of this into account, Logistic Regression is recommended as the most dependable model for this dataset. We are confident that our acquired analysis and result is correct.